

# Dynamic FIDVR Response in Transmission-Distribution Co-Simulation

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**Abstract**—Fault induced delayed voltage recovery (FIDVR) is a post-fault voltage phenomenon mainly driven by the stalling of single-phase induction motors, in particular residential air-conditioner compressor motors. The literature on FIDVR can be broadly divided into three groups. Positive-sequence composite load models are computationally efficient and are widely used in planning studies, but they represent the feeder as an equivalent load. Electromagnetic-transient(EMT) and hybrid EMT transient stability models provide a detailed representation of motor and phase domain dynamics, but their computational burden limits their use in large system studies. Transmission-distribution (T&D) co-simulation provides an intermediate solution, since it preserves the transmission dynamic model and the feeder-level voltage response in separate, coordinated simulators. This report reviews these approaches and motivates a feeder-level FIDVR assessment using an open-source T&D co-simulation framework. The proposed study couples an IEEE 14-bus transmission system and an IEEE 13-node distribution feeder through ANDES-OpenDSS-HELICS. A three-phase fault is applied at the interface bus, and selected single-phase feeder loads are represented by Motor D-type compressor behavior. The objective is to quantify the post-fault voltage recovery trajectory and to further determine the mitigation methods for FIDVR studies.

**Index Terms**—fault induced delayed voltage recovery, transmission-distribution co-simulation, dynamic load modelling, HELICS, OpenDSS, ANDES, composite load model.

## I. INTRODUCTION

### A. Motivation

Power system dynamic studies have traditionally separated transmission and distribution models. This separation is justified in many applications because transmission networks are usually modelled as balanced positive-sequence systems, whereas distribution networks are radial, unbalanced, and rich in device-level controls. However, the increasing interaction between bulk-system disturbances and distribution-side dynamics makes this separation less accurate for some voltage-recovery phenomena [1], [2]. Fault-induced delayed voltage recovery (FIDVR) is one such phenomenon. It is characterised by a sustained low-voltage profile following the clearing of a transmission fault, and several FIDVR events have been reported in North American systems [3], [4].

A FIDVR event is initiated by a fault. After the fault is cleared, the voltage first recovers only partially and can remain depressed for several seconds. The North American Electric Reliability Corporation (NERC) describes FIDVR as a condition involving induction motor stalling, incomplete initial

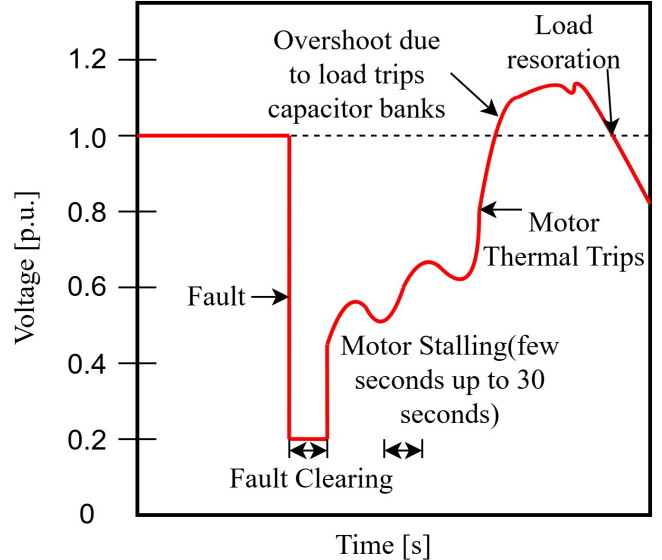


Fig. 1. Representative FIDVR voltage recovery trajectory [7].

voltage recovery, and delayed restoration of the voltage toward its pre-contingency level [5], [6]. Figure 1 illustrates the typical voltage trajectory of a FIDVR event. The dominant physical mechanism is the stalling of single-phase induction motors in residential air conditioners. During a severe voltage sag, stalled compressor motors draw a large reactive current. This current further depresses the local voltage, delays motor recovery, and can lead to contactor dropout, thermal tripping, or additional motor stalling. The resulting feedback between load behaviour and voltage recovery is the main reason why FIDVR cannot always be represented accurately by a static load model.

This report reviews the modeling approaches used for FIDVR analysis and develops a research framework for dynamic FIDVR response in T&D co-simulation. Section II reviews motor-level FIDVR models, aggregate composite load models, and T&D co-simulation methods. Section III presents the proposed ANDES-OpenDSS-HELICS framework, the Motor D parameterisation, and the assessment metrics. Section IV summarises the expected contribution.

## II. LITERATURE REVIEW

### A. FIDVR Mechanism and Load Modeling

Residential air-conditioning load is a major driver of FIDVR. A U.S. Department of Energy survey reported that a large share of households use either central or room air-conditioning equipment; consequently, faults in areas with high air-conditioner penetration can lead to widespread motor stalling [8]. Therefore, an adequate representation of single-phase A/C motors is essential for reproducing the delayed voltage-recovery process.

Several studies have modeled the compressor motor explicitly. Ravikumar *et al.* developed an EMTP-based point-on-wave model of single-phase air-conditioner motors and tested the response under different voltage-depression conditions [9]. Zheng and DeMarco proposed a dynamic performance model that represents the transition from running to stalled operation and the subsequent thermal trip process [10]. These studies show that FIDVR is not determined only by fault duration or voltage magnitude. It is also affected by motor loading, protection settings, contactor behavior, and thermal trajectories.

A second approach is to represent the phenomenon through aggregate dynamic load models. The WECC composite load model includes a dedicated single-phase air-conditioner motor component and associated protection logic [11]. The NERC dynamic load modeling reference provides practical default parameters for the single-phase motor component commonly referred to as Motor D [5]. Motor D model provides a compact performance-based representation of compressor motor behavior during voltage sags. In the original formulation [11], a prescribed fraction of the constant power load is reassigned to an equivalent air-conditioner motor, while the remaining part of the load is represented by the associated static load model. This representation retains the main FIDVR mechanisms of residential compressor motors, including voltage-dependent stalling, stalled-state active and reactive power consumption, contactor action, thermal protection, and partial restart. It is computationally simpler than an EMT motor model and is therefore suitable for feeder-level dynamic studies within a T&D co-simulation framework. Its limitation is that the motor is represented by a performance model rather than by instantaneous electromagnetic equations. Consequently, point-on-wave effects, dc offsets, and detailed flux dynamics are not resolved. These effects require EMT modelling, but at a substantially higher computational cost and with a smaller practical study area [11]–[13].

Composite load models and EMT models are based on different modelling assumptions and are therefore suited to different studies. The composite load model is an aggregate phasor-domain model. It is computationally efficient and well suited to large-scale dynamic simulations, but the single-phase compressor motor is described through an equivalent performance model rather than through instantaneous electromagnetic variables. Conversely, an EMT model resolves the point-on-wave evolution of voltages, currents, flux linkages, and

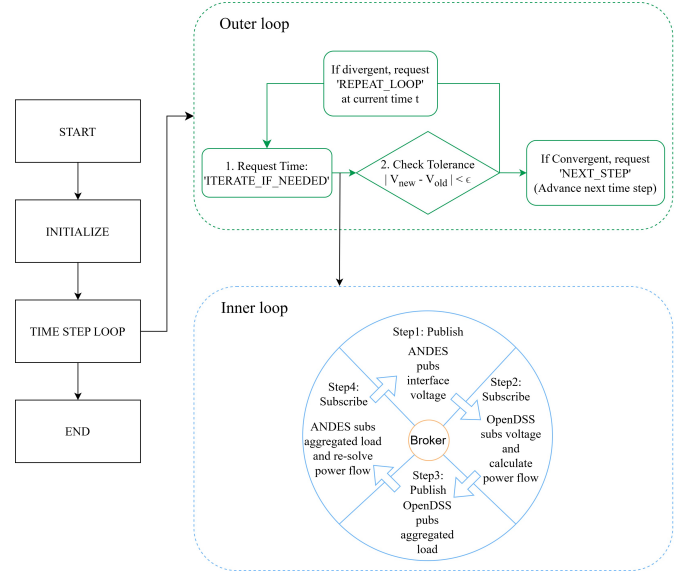


Fig. 2. Iterative time-advancement structure of the HELICS-based co-simulation.

torque. It can therefore represent phase-angle dependence, dc-offset components, and unbalanced waveform effects that are not retained in a positive-sequence formulation. This increased physical resolution, however, is obtained at the expense of a substantially higher computational cost and a reduced practical study area [12], [13].

### B. Transmission-Distribution Co-Simulation

T&D co-simulation keeps the transmission and distribution models in separate simulation tools while coordinating time advancement and exchanging boundary variables through middleware. In a typical dynamic T&D interface, the transmission simulator sends the interface-bus voltage to the distribution simulator, and the distribution simulator returns the complex feeder power to the transmission simulator. This structure allows each subsystem to retain its native models and numerical solution methods while appearing externally as a coupled system. [14]

Huang *et al.* proposed an open-source framework for power-system T&D dynamic co-simulation based on the Framework for Network Co-Simulation (FNCS) [15]. Palmintier *et al.* later described the HELICS architecture, which was developed to coordinate large-scale co-simulation studies involving power, communication, market, and end-use simulators [16]. HELICS also supports iterative time advancement. As illustrated in Fig. 2, the outer loop advances the global simulation time, whereas the inner loop exchanges boundary variables among federates at a fixed time instant. If the boundary mismatch is above the convergence tolerance, the federates remain in the ITERATING state. Once the boundary variables are consistent, HELICS grants the next time step and the co-simulation proceeds.

Different tool combinations have been used for T&D co-simulation. Early steady-state and quasi-static studies coupled

MATLAB with OpenDSS for IEEE 9-bus or IEEE 39-bus transmission systems and EPRI distribution feeders [17], [18]. Other studies used MATLAB-based transmission and distribution models for coordinated T&D optimal power flow [19]. Integrated grid studies coupled FESTIV or MATPOWER with GridLAB-D to connect bulk-system operation with distribution and end-use models [20]. For dynamic studies, researchers have used PSCAD/EMTDC with InterPSS for hybrid EMT-phasor simulation, mixed three-sequence and three-phase models for unbalanced T&D dynamics, Three Phase Dynamics Analyzer for combined T&D dynamic simulation, and GridLAB-D or EMT/OpenDSS for distribution-side dynamic studies [21]–[25]. Federated dynamic co-simulation has also been studied using FNCS with dynamic three-phase models and GridLAB-D, PSAT with OpenDSS for integration time step analysis, and PSLF with OpenDSS for distributed generation response to bulk system faults [15], [26].

More recently, Wang *et al.* developed an open-source ANDES-OpenDSS-HELICS framework for DER frequency-response studies [14]. In that framework, ANDES represents the transmission dynamics, OpenDSS solves the distribution network, and HELICS coordinates the exchange of boundary quantities. The framework is particularly relevant because it demonstrates the feasibility of an open-source and scalable T&D dynamic co-simulation architecture. However, its main objective is DER frequency response rather than FIDVR. The present work uses the same general software philosophy but focuses on feeder-level compressor-motor stalling and delayed voltage recovery.

### C. Position of the Proposed Work

Table I summarizes the main modeling approaches. Existing FIDVR studies have established the physical basis of delayed voltage recovery, the importance of stalled residential compressor motors, and the usefulness of aggregate composite load models. EMT and hybrid EMT transient-stability studies provide additional physical detail but are often limited to simplified feeders or reduced study areas. Recent T&D co-simulation studies demonstrate that distribution-side behavior can modify bulk-system dynamics, but many of them focus on DER ride-through rather than compressor-motor stalling. The proposed work therefore targets the intermediate gap: an open-source, feeder-level, FIDVR-oriented dynamic co-simulation study using ANDES, OpenDSS, and HELICS.

## III. MODELING FRAMEWORK

### A. General Dynamic Representation

The transmission system is represented by the following set of differential-algebraic equations (DAEs) [32]:

$$\dot{x}_T = f_T(x_T, y_T, u_T, p_T), \quad (1)$$

$$0 = g_T(x_T, y_T, u_T, p_T), \quad (2)$$

where  $x_T$  denotes the transmission dynamic states,  $y_T$  is the algebraic variables,  $u_T$  is the boundary input received from the distribution feeder, and  $p_T$  is the model parameters.

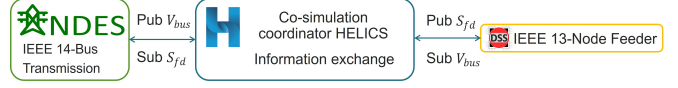


Fig. 3. Proposed ANDES-OpenDSS-HELICS co-simulation framework.

For feeder  $k$ , the distribution model is written as

$$\dot{x}_{D,k} = f_{D,k}(x_{D,k}, y_{D,k}, u_{D,k}, p_{D,k}), \quad (3)$$

$$0 = g_{D,k}(x_{D,k}, y_{D,k}, u_{D,k}, p_{D,k}). \quad (4)$$

For a quasi-static OpenDSS feeder, the differential states may be absent from the network solution itself. In this study, motor stalling, contactor dropout, thermal tripping, and restart logic are represented as discrete dynamic states in the Python layer surrounding OpenDSS.

At the interface bus, the transmission simulator sends the feeder-source voltage and the distribution simulator returns the aggregate feeder power. The exchanged variables can be written as

$$u_{D,k} = V_{abc,k}, \quad (5)$$

$$u_T = \sum_{k=1}^{N_f} S_{abc,k}, \quad (6)$$

where  $V_{abc,k}$  is the three-phase interface voltage supplied to feeder  $k$ ,  $S_{abc,k}$  is the three-phase complex power drawn by feeder  $k$ , and  $N_f$  is the number of coupled feeders.

### B. Test System and Event Definition

The proposed assessment uses the IEEE 14-bus transmission system coupled to the IEEE 13-node feeder at bus 14. ANDES solves the transmission dynamic model, OpenDSS solves the distribution feeder, and HELICS coordinates the boundary-variable exchange. The transmission federate publishes the interface-bus voltage, denoted by `TxInterfaceVoltage`. The distribution federate publishes the aggregate feeder complex power, denoted by `Feeder_Power`. The overall structure is shown in Fig. 3.

A three-phase-to-ground fault is applied at the interface bus at  $t = 1.0$  s. The default fault duration is 0.08 s. The default fault resistance is zero, and the fault reactance is 0.3 p.u. The transmission time-domain step is set to 0.03 s, which is also used as the default co-simulation step in the transmission federate [33]. Before the dynamic simulation starts, the transmission and distribution simulators iterate on the interface voltage and feeder power to obtain a consistent initial operating point.

On the distribution side, Bus 675 is selected as the default monitored load bus. OpenDSS controls are disabled so that the FIDVR control logic is explicitly represented in the Python layer. The feeder source voltage is initialized at 1.03 p.u. before the co-simulation initialization.

The added compressor motor groups are placed at selected single-phase load locations of the IEEE 13-node feeder, as shown in Fig. 4:

$$\{634(a, b, c), 645, 675(a, b, c), 611, 652, 670(a, b, c)\}.$$

TABLE I  
COMPARISON OF AVAILABLE APPROACHES FOR SIMULATING FIDVR

| Approach                                                              | Main contribution                                                                                                                                                  | Network representation                                                             | Remaining limitation                                                                                                              |
|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Field motivated FIDVR models [3]                                      | Combined field observations and laboratory tests to explain delayed voltage recovery and to motivate dynamic load representation.                                  | 115 kV and 66 kV sub-transmission systems.                                         | Limited feeder level and phase domain resolution.                                                                                 |
| WECC composite load model [27]                                        | Implemented and calibrated aggregate composite load models, compared simulated recovery with PMU data, and identified sensitive Motor D and protection parameters. | Southern California Edison Valley substation.                                      | Does not resolve feeder topology, phase imbalance, or the spatial distribution of stalled single-phase motors.                    |
| WECC CMPLDW with variable-frequency drive (VFD) air conditioners [28] | Shown that higher VFD penetration can reduce voltage sag and recovery time.                                                                                        | Utility planning network.                                                          | Focuses on the effect of VFD A/C penetration rather than feeder level motor stalling diversity.                                   |
| EMTP point-on-wave model [29]                                         | Demonstrated that motor stalling depends on voltage magnitude, duration, loading condition, and point-on-wave initiation angle.                                    | Simplified unbalanced distribution feeder.                                         | High model detail limits scalability for large system dynamic studies.                                                            |
| Hybrid EMT transient stability model [30]                             | Coupled PSCAD/EMTDC and InterPSS through a three-phase boundary equivalent to study unbalanced faults and A/C motor stalling in a large WECC system.               | Sub-transmission and distribution systems supplied from selected WECC buses.       | Uses equivalent feeders and a fundamental frequency interface, which limits detailed feeder topology and harmonic representation. |
| Coupled PSLF-OpenDSS study with WECC CMPLDWG and PVD1 [31]            | Shown that feeder level inverter ride-through behavior can significantly change the bulk system response during a disturbance.                                     | WECC transmission case with distribution feeders and IEEE 1547 ride-through logic. | Focuses on DER/PV ride-through rather than compressor-motor stalling and FIDVR motor dynamics.                                    |
| Proposed work                                                         | Develops an open-source, feeder-level, FIDVR oriented extension of T&D dynamic co-simulation using ANDES-OpenDSS-HELICS.                                           | IEEE 14-bus transmission system coupled to an IEEE 13-node feeder.                 |                                                                                                                                   |

TABLE II  
BASIC INFORMATION OF THE IEEE 13-NODE FEEDER USED IN THE PROPOSED STUDY

| Item                        | Value                        |
|-----------------------------|------------------------------|
| Total active load           | 3466 kW                      |
| Total reactive load         | 2102 kvar                    |
| Capacitor at Bus 675        | 600 kvar, three-phase        |
| Capacitor at Bus 611        | 100 kvar, single-phase       |
| Maximum Motor D active load | 1911 kW                      |
| Motor D active load share   | 55.1%                        |
| Motor locations             | 634, 645, 675, 611, 652, 670 |

For each selected phase load, the original load is divided into a static component and a motor component. The maximum motor share is defined by replacing all original static active loads at these selected locations with the Motor D compressor load, which is

$$P_{\text{MotorD,max}} = 1911 \text{ kW}. \quad (7)$$

Compared with the total feeder active load,

$$P_{\text{load}} = 3466 \text{ kW}, \quad (8)$$

the maximum Motor D active-load share is

$$\frac{P_{\text{MotorD,max}}}{P_{\text{load}}} = \frac{1911}{3466} = 55.1\%. \quad (9)$$

Thus, in the high motor shared case, more than half of the feeder active load is represented by Motor D compressor load. The corresponding reactive power will be determined by the motor equations, terminal voltage, and motor state during

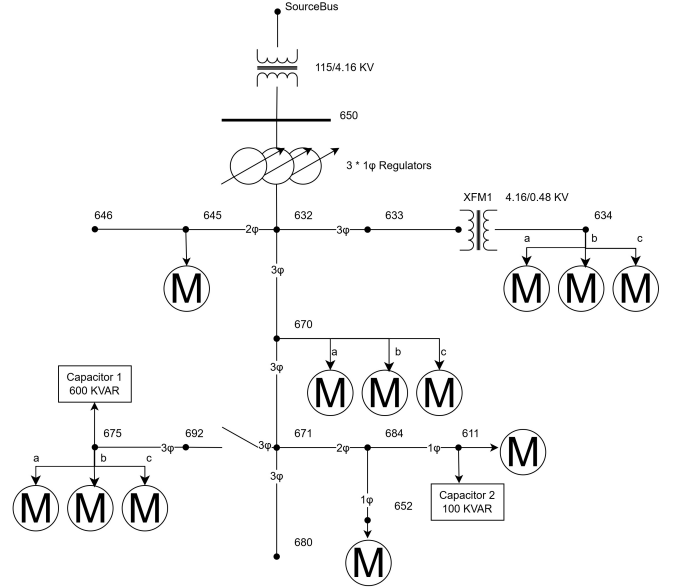


Fig. 4. IEEE 13-node feeder with added compressor-motor groups.

the simulation. The basic feeder and motor-share information is summarized in Table II.

### C. Motor D Parameterization

The compressor-motor representation follows the WECC single-phase air-conditioner Motor D structure [11]. For each

selected phase load  $i$ , the original active power  $P_{0,i}$  is split as

$$P_{m,i} = F_m P_{0,i}, \quad (10)$$

$$P_{s,i} = (1 - F_m) P_{0,i}, \quad (11)$$

where  $P_{m,i}$  is the motor component,  $P_{s,i}$  is the remaining static component, and  $F_m$  is the motor fraction. The initial motor reactive power is computed from the compressor power factor as

$$Q_{m,i} = P_{m,i} \tan [\cos^{-1}(\text{CompPF})]. \quad (12)$$

For the default value  $\text{CompPF} = 0.97$ , the initial reactive-to-active power ratio is

$$\frac{Q_{m,i}}{P_{m,i}} = 0.251. \quad (13)$$

Thus, before stalling, the motor component has a moderate reactive-power demand.

During the stalled condition, the Motor D implementation uses the stall impedance parameters. The equivalent conductance and susceptance are

$$G_{stall} = \frac{R_{stall}}{R_{stall}^2 + X_{stall}^2}, \quad (14)$$

$$B_{stall} = \frac{X_{stall}}{R_{stall}^2 + X_{stall}^2}, \quad (15)$$

and the corresponding stalled active and reactive powers are

$$P_{stall} = G_{stall} V^2, \quad (16)$$

$$Q_{stall} = B_{stall} V^2. \quad (17)$$

Using  $R_{stall} = 0.124$ ,  $X_{stall} = 0.114$ , and  $V_{stall} = 0.70$ , the stalled load factors at the stall threshold are approximately

$$P_{stall} \approx 2.14 P_m, \quad Q_{stall} \approx 1.97 P_m. \quad (18)$$

The stalled motors therefore impose a large reactive power demand on the feeder and can delay the restoration of the post-fault voltage. The default parameters used in the implemented feeder model are summarized in Table III.

#### D. FIDVR Assessment Metrics

The assessment compares two main cases:

- 1) **Baseline co-simulation case:** the IEEE 13-node feeder is coupled to the transmission system without the added single-phase compressor-motor groups.
- 2) **FIDVR-enabled case:** the compressor-motor groups are added with different motor fractions to evaluate the sensitivity of post-fault recovery to motor penetration.

TABLE III  
DEFAULT MOTOR D PARAMETERS USED IN THE FEEDER MODEL

| Parameter    | Value      | Meaning                               |
|--------------|------------|---------------------------------------|
| CompPF       | 0.97       | Compressor operating power factor     |
| $V_{stall}$  | 0.70 p.u.  | Stall voltage threshold               |
| $R_{stall}$  | 0.124 p.u. | Stall resistance                      |
| $X_{stall}$  | 0.114 p.u. | Stall reactance                       |
| $T_{stall}$  | 0.033 s    | Stall delay                           |
| $F_{rst}$    | 0.20       | Fraction allowed to restart           |
| $V_{rst}$    | 0.90 p.u.  | Restart voltage threshold             |
| $T_{rst}$    | 0.40 s     | Restart delay                         |
| $V_{brk}$    | 0.86 p.u.  | Running-characteristic break voltage  |
| $V_{c1,off}$ | 0.45 p.u.  | Contact dropout begins                |
| $V_{c2,off}$ | 0.35 p.u.  | Contact dropout completes             |
| $V_{c1,on}$  | 0.50 p.u.  | Contact reconnection completes        |
| $V_{c2,on}$  | 0.40 p.u.  | Contact reconnection begins           |
| $T_{th}$     | 10.0 s     | Thermal time constant                 |
| $Th1t$       | 1.30       | Thermal trip begins                   |
| $Th2t$       | 4.30       | Thermal trip completes                |
| $F_{uvr}$    | 0.00       | Fraction with undervoltage relay trip |
| $V_{uv1}$    | 0.80 p.u.  | First UV trip threshold               |
| $T_{uv1}$    | 0.20 s     | First UV trip delay                   |
| $V_{uv2}$    | 0.90 p.u.  | Second UV trip threshold              |
| $T_{uv2}$    | 5.00 s     | Second UV trip delay                  |

The following quantities are used to evaluate the response:

$$V_{min}^T = \min_t |V_T(t)|, \quad (19)$$

$$V_{min}^{fh} = \min_t |V_{fh}(t)|, \quad (20)$$

$$V_{min}^{load} = \min_{\ell,t} |V_{\ell}(t)|, \quad (21)$$

$$Q_{max}^D = \max_t Q_D(t), \quad (22)$$

$$T_{0.90}^D = \min\{t > t_c : |V_D(t)| \geq 0.90\}, \quad (23)$$

$$T_{0.95}^D = \min\{t > t_c : |V_D(t)| \geq 0.95\}, \quad (24)$$

$$N_{stall}(t) = \text{number of stalled motor groups.} \quad (25)$$

Here,  $V_T$  is the interface bus voltage,  $V_{fh}$  is the feeder-head voltage,  $V_{load}$  is the voltage of monitored load bus  $\ell$ ,  $Q_D$  is the feeder reactive power seen by the transmission system, and  $t_c$  is the fault clearing time. The recovery times  $T_{0.90}^D$  and  $T_{0.95}^D$  directly measure the delayed voltage recovery process.

The comparison is performed between two T&D co-simulation cases. The baseline case uses the original feeder loads without added compressor motors. The FIDVR-enabled case uses the same system and disturbance, but selected single-phase loads are split into a residual static component and a WECC compressor-motor component. Therefore, the difference between the two cases isolates the effect of compressor-motor stalling.

The motor-induced voltage depression is defined as

$$\Delta V_{min}^{fh} = V_{min}^{fh,base} - V_{min}^{fh,motor}, \quad (26)$$

$$\Delta V_{min}^{load} = V_{min}^{load,base} - V_{min}^{load,motor}. \quad (27)$$

Positive values indicate that the compressor-motor component deepens the post-fault voltage sag. The corresponding recovery-time delay is defined as

$$\Delta T_{\alpha} = T_{\alpha}^{motor} - T_{\alpha}^{base}, \quad (28)$$

where

$$T_{\alpha}^c = \min\{t > t_c : |V_{load}^c(t)| \geq \alpha\}, \quad c \in \{\text{base}, \text{motor}\}. \quad (29)$$

Here,  $t_c$  is the fault clearing time and  $\alpha$  is the selected recovery threshold, such as 0.90 or 0.95 p.u. A larger  $\Delta T_{\alpha}$  indicates that compressor-motor stalling delays the post-fault voltage recovery. The increase in feeder reactive power demand is also quantified as

$$\Delta Q_{\max}^D = Q_{\max}^{D, \text{motor}} - Q_{\max}^{D, \text{base}}. \quad (30)$$

#### IV. CASE STUDIES

Firstly, to illustrate the effectiveness of the selected co-simulation framework, a validation system of the IEEE 118-bus transmission system and IEEE 8500-nodes feeder for in ANDES-OpenDSS-HELICS is compared with GridPACK-GridLAB-D-HELICS framework. After the validation case, the proposed co-simulation connects the IEEE 14-bus transmission system with IEEE 13-nodes test feeder. Various motor share and faults are tested to illustrate the FIDVR trajectory. The testing simulations are performed on a personal computer with intel CORE i7 as the CPU.

A. Validation: IEEE 118-bus and IEEE 8500-nodes system

B. Proposed framework: IEEE 13-bus and IEEE 14-nodes system

#### V. CONCLUSION

This report reviews the main modeling approaches used for FIDVR analysis. Field and laboratory studies established the link between delayed voltage recovery and stalled single-phase air-conditioner compressor motors. Positive-sequence composite load models, especially the WECC composite load model, provide a practical representation for transmission planning studies. However, these models aggregate feeder behavior into an equivalent load and cannot directly represent feeder voltage diversity, load bus voltage depression, or phase level differences. EMT and hybrid EMT-transient stability studies improve the physical detail of the motor and fault response, but they often require simplified feeders or reduced study areas. Recent T&D co-simulation studies show that distribution-side behavior can affect bulk-system dynamics, but most of them focus on DER ride-through rather than compressor-motor stalling.

The proposed work develops a FIDVR-oriented T&D co-simulation study using ANDES-OpenDSS-HELICS. The objective is to place the Motor D-typed compressor behavior at selected feeder loads and quantify its impact on post-fault voltage recovery. The study compares a co-simulation baseline without added compressor motors against FIDVR-enabled cases with different motor fractions. The expected contribution is an open-source, feeder level workflow for assessing FIDVR in dynamic T&D co-simulation. Future work should test sensitivity to fault severity and motor share and extend the study from the IEEE 13-node feeder to larger and more unbalanced distribution feeders.

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